MTH 107(7 and 10) Dr. James Ashe
**Exam 3 Review** Exam 3 is on theMonday $27^{\text{th}}$ October, 2014

**1.** A card is dealt from a well-shuffled, standard deck of 52 cards. Find the probability of being dealt each of the following.

**a.** A red card
$$\frac{n(\text{red cards})}{n(\text{cards})} = \frac{26}{52} = \frac{1}{2}$$

**b.** A queen
$$\frac{n(\text{queens})}{n(\text{cards})} = \frac{4}{52} = \frac{1}{13}$$

**c.** A club
$$\frac{n(\text{clubs})}{n(\text{cards})} = \frac{13}{52} = \frac{1}{4}$$

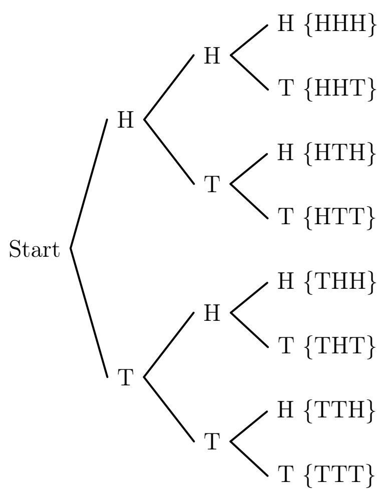
**d.** The queen of clubs
$$\frac{n(\text{queen of clubs})}{n(\text{cards})} = \frac{1}{52}$$

**e.** A queen or a club
$$\frac{n(\text{queens or clubs})}{n(\text{cards})} = \frac{4+13-1}{52} = \frac{16}{52} = \frac{4}{13}$$

**f.** not a queen
$$1 - \frac{4}{52} = \frac{48}{52} = \frac{12}{13}$$

**2.** Three coins are tossed. Find each of the following:

**a.** the sample space




**b.** the event, $E_1$, that exactly two are tails
$$E_1 = \{HTT, THT, TTH\}$$

**c.** the event, $E_2$, that at least two are tails.
$$E_2 = \{HTT, THT, TTH, TTT\}$$

**d.** the probability and odds of $E_1$
$$p(E_1) = \frac{3}{8}$$
$$o(E_1) = 3 : 5$$

**e.** the probability and odds of $E_2$
$$p(E_2) = \frac{4}{8} = \frac{1}{2}$$
$$o(E_2) = 1 : 1$$

**3.** Five cards are dealt (without replacement) from a well-shuffled standard deck. **Just write the formula** for the probability of the following:

**a.** being dealt 5 spades
$$\frac{{}_{13}C_5}{{}_{52}C_5}$$

**b.** being dealt 4 spades and 1 heart
$$\frac{{}_{13}C_4 \cdot {}_{13}C_1}{{}_{52}C_5} = \frac{{}_{13}C_4 \cdot 13}{{}_{52}C_5}$$

**c.** being dealt all the same suit.
$$\frac{{}_{4}C_1 \cdot {}_{13}C_5}{{}_{52}C_5} = \frac{4 \cdot {}_{13}C_5}{{}_{52}C_5}$$

**d.** not being dealt all the same suit.
$$1 - \frac{4 \cdot {}_{13}C_5}{{}_{52}C_5}$$

**4.** A drawer has 7 socks. 4 socks are black and 3 are white socks. John randomly pulls out 4 socks. Find the probability of the following:

**a.** all 4 socks are black.

$$\frac{{}_4C_4}{{}_7C_4} = \frac{1}{35}$$

**b.** exactly 2 are white.

$$\frac{{}_3C_2 \cdot {}_4C_2}{{}_7C_4} = \frac{3 \cdot 6}{35} = \frac{18}{35}$$

2 are white and 2 are black

**c.** at least 3 are white.

$$\frac{{}_3C_3 \cdot {}_4C_0 + {}_3C_4 \cdot {}_4C_0}{{}_7C_4} = \frac{1 \cdot 1 + 0 \cdot 1}{35} = \frac{1}{35}$$

3 OR 4 are white

**d.** at most 2 are black.

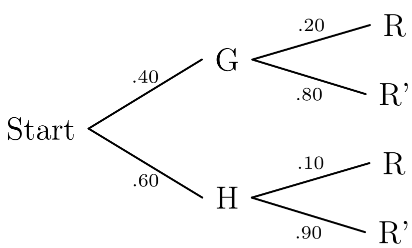
$$\frac{{}_4C_0 \cdot {}_3C_4 + {}_4C_1 \cdot {}_3C_3 + {}_4C_2 \cdot {}_3C_2}{{}_7C_4} = \frac{1 \cdot 0 + 4 \cdot 1 + 6 \cdot 3}{35} = \frac{22}{35}$$

0 or 1 or 2 are black.

- Note: Above, ${}_3C_4 = 0$ as there are 0 ways to choose 4 white socks if there are only 3. I will **not** have a situation like this on the exam.

**5.** Gregor's Garden Corner buys 40% of its plants from the Green Growery and the rest from Herb's Herbs. Of the plants from the Green Growery, 20% must be returned, and 10% of those from Herb's Herbs must be returned.

**a.** draw a tree diagram describing the probability




**b.** find the probability that a plant must be returned and it is from Herb's Herbs.

$p(H \cap R) = (.60)(.10) = .06$

**c.** find the probability that a plant from the Green Growery must be returned.

$p(R \mid G) = .20$

**d.** find the probability that a plant must be returned.

A returned plant can be from Herb's OR Gregory,

$p(R) = p(H \cap R) + p(G \cap R) = (.6)(.1) + (.4)(.2) = .06 + .08 = .14$

**6.** Use the provided table to answer the following questions.

| **Age** | **Obama** | **McCain** | **Other/NA** |
|---|---|---|---|
| 18-29 | 11.9% | 5.8% | 0.4% |
| 30-44 | 15.1% | 13.3% | 0.6% |
| 45-64 | 18.5% | 18.1% | 0.4% |
| 65+ | 7.2% | 8.5% | 0.3% |

**a.** What is the probability that a voter under 45 voted for McCain.

$p(M \mid \text{under } 45) = \dfrac{n(M \cap \text{under } 45)}{n(\text{under } 45)} = \dfrac{.058 + .133}{.119 + .058 + .004 + .151 + .133 + .006} = \dfrac{.191}{.471} \approx 40.55\%$

**b.** What is the probability that a person who voted for McCain is 45 or older.

$p(\text{over } 45 \mid M) = \dfrac{n(\text{over } 45 \cap M)}{n(M)} = \dfrac{.181 + .085}{.058 + .133 + .181 + .085} = \dfrac{.266}{.457} \approx 58.21\%$

**7.** MHU is holding a raffle to raise money for a new internet service. The tickets cost \$10 each. The value and number of each prize to be given away is listed in the provided table. Find the expected value of a ticket under the given circumstances.

| **Prize** | **Value** | # |
|---|---|---|
| Vacation | \$3,000 | 1 |
| Stereo | \$500 | 5 |
| Books | \$100 | 10 |
| T-shirt | \$20 | 100 |

**a.** 200 tickets

There are $200 - 116 = 84$ losing tickets.

$EV = 2990(\frac{1}{200}) + 490(\frac{5}{200}) + 90(\frac{10}{200}) + 10(\frac{100}{200}) - 10(\frac{84}{200})$
$= \$32.5$

**b.** 1000 tickets

There are $1000 - 116 = 884$ losing tickets.

$EV = 2990(\frac{1}{1000}) + 490(\frac{5}{1000}) + 90(\frac{10}{1000}) + 10(\frac{100}{1000}) - 10(\frac{884}{1000})$
$= -\$1.5$

**8.** Of all the flashlights is a large shipment, 15% have a defective bulb, 10% have a defective battery, and 5% have both defects. If you buy one of these flashlights, find the probability of the following:

**a.** a defective bulb or a defective battery

$p(Batt' \cup Bulb') = .10 + .05 + .05$
$= .20$

**b.** a good bulb or a good battery

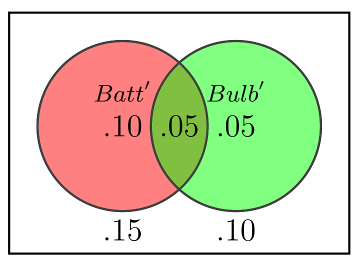
$p((Batt' \cap Bulb')) = 1 - .05 = .95$

**c.** a defective bulb and a defective battery.

$p(Bulb' \cap Batt') = .05$

**d.** just a defective bulb

$p(Bulb' - Batt') = .10 - .05 = .05$



**9.** A single card is dealt from a well-shuffled standard deck. The probability of being dealt a jack is $4/52$, and the probability of being dealt a red card is $1/2$.

**a.** Are the events "being dealt a jack" and "being dealt a red card" independent?

$$p(J) = \frac{4}{52} = \frac{1}{13} \text{ and } p(J \mid R) = \frac{n(J \cap R)}{n(R)} = \frac{2}{26} = \frac{1}{13}$$

So yes they are indepedent since $p(J) = p(J \mid R)$.

**b.** Are the events "being dealt a jack" and "being dealt a red card" mutually exclusive?

$J \cap R \neq \emptyset$, since there is a red jack (the jack of hearts and the jack of diamonds). So they are not mutually exclusive.

**c.** Are the events "being dealt a jack" and "being dealt a queen" mutually exclusive?

$J \cap Q = \emptyset$, since a card cannot be both a queen and jack.
So they are mutually exclusive.

**d.** Suppose instead that a card is dealt, replaced, and then another card is dealt. Are the events "being dealt a jack 1$^{\text{st}}$" and "being dealt a queen 2$^{\text{nd}}$ independent? are they mutually exclusive?

Yes they are independent. Since the card was replaced and reshuffled, the 1$^{\text{st}}$ draw does not affect the probability of the 2$^{\text{nd}}$ draw. They are not mutually exclusive because you can get a Jack and then a Queen.

**10.** A fair six-sided dice is rolled 5 times. Find the probability of the following:

**a.** All 2's are rolled.

$$p(\text{five } 2's) = \frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{6^5}$$

**b.** Four 2's are rolled and the 5$^{\text{th}}$ roll is a 1.

$$p(\text{four } 2's \cap \text{one } 1) = (\frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{6}) \cdot \frac{1}{6} = \frac{1}{6^5}$$